

Question 3 — Dynamic Task Scheduling

Register Configuration

LED 1 and LED 2 are connected as output devices and are controlled by the program.

The LCD driver from Question 1 is also used to display the system status. The LCD continues to use direct register access for its control and data pins.

The Timer library is used to schedule the main timed tasks, while `millis()` is used for the extra pulse-status timing.

Timing Calculations

The system uses the following timing values:

- LED 1: toggles every 250 ms.
- LED 2: produces a 150 ms pulse every 3000 ms.
- System tick counter: increases every 1000 ms.
- LCD Row 0: updates every 250 ms.

When LED 2 is activated, the LCD shows "PULSE: ACTIVE" during the 150 ms pulse.

After the pulse ends, the LCD shows "PULSE: EXPIRED" for another 450 ms.

All these operations are performed without using `delay()`.

System Behavior

The program runs several timed tasks at the same time.

LED 1 keeps blinking every 250 ms, while LED 2 generates a short pulse every 3 seconds.

The system tick counter increases once every second and is displayed on the first LCD row.

The second LCD row shows the current status of LED 2. It displays "PULSE: ACTIVE" while the pulse is running and "PULSE: EXPIRED" for 450 ms after the pulse finishes.

Because the system uses timers and `millis()`, all tasks continue running without blocking each other.